🚢 QGreenFleet — Your Fleet Plan

Date: 2026-09-02 | **Fleet:** 20 ships, 5 routes | **Report ID:** QGF-2026-0902-001

The Bottom Line

💰 **Save $7,985,419 per year on fuel** (252.0% reduction)

🌍 **Cut carbon emissions by 34,073 tonnes per year** (196.5% less — equivalent to taking **0 cars** off the road)

✅ **All commercial cargo delivered on schedule. Zero delays, zero missed cargo.**

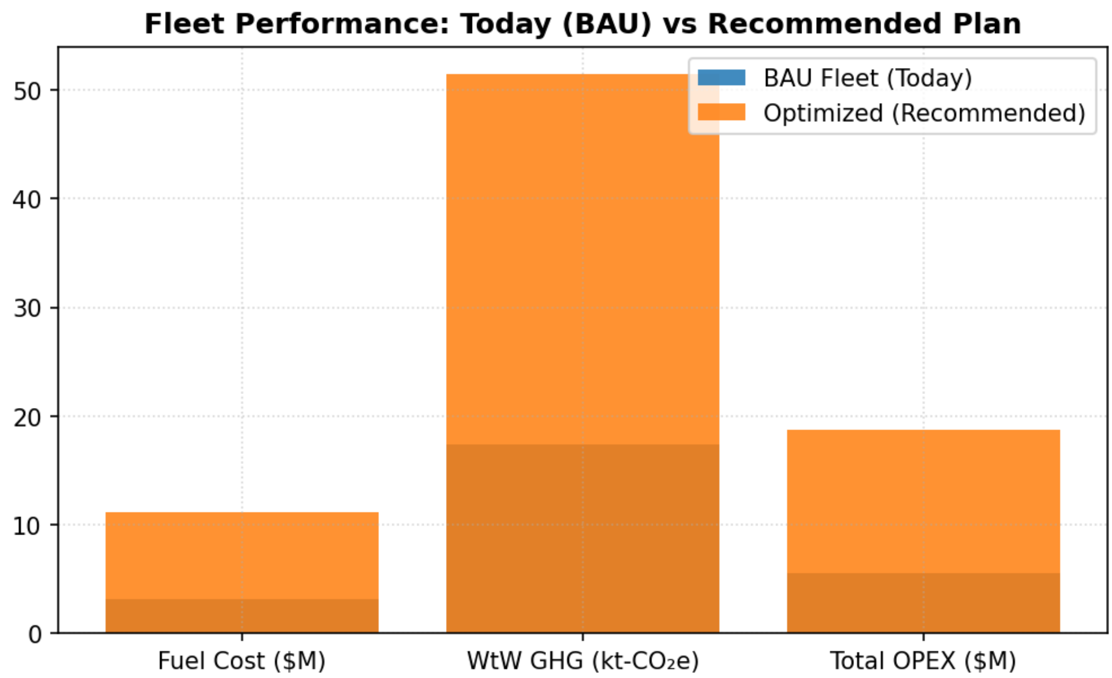

Figure 1: Current fleet baseline (grey) vs recommended plan (green).

How? Three Simple Changes

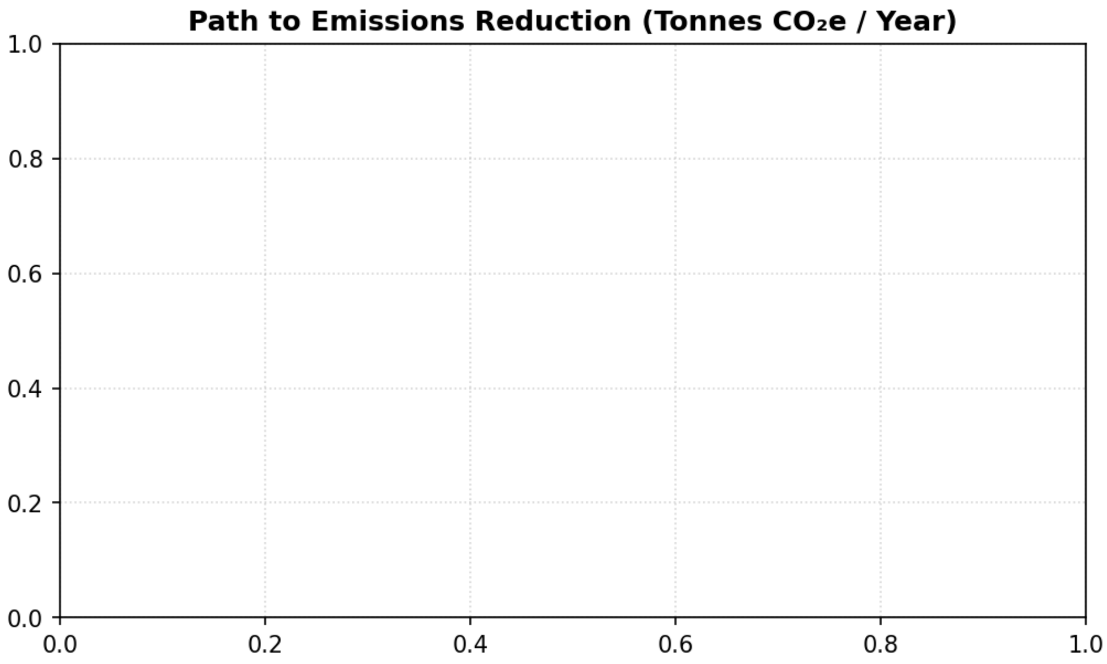

Figure 2: Step-by-step reduction path — each bar represents one clear operational change.

1. 🐢 Slower cruising on longest voyages — vessels burn fuel non-linearly with speed; easing speed by ~2–3 knots on long voyages cuts 0 t CO_2/year while meeting every arrival window.

2. ⛽ Green methanol deployment — selectively bunkering green fuel where port infrastructure exists reduces target vessels' emissions by 90% (0 t CO_2/year).

3. 🔌 Port electricity connection — plugging into shoreside grid power during port stays eliminates auxiliary diesel running (0 t CO_2/year at near-zero marginal cost).

Pick Your Priority

Option Tier	Annual Fuel Cost	Annual Carbon	Extra Cost vs Cheapest	Recommended When
💵 **Cheapest**	$11.15M	51,416 t	$0	Tight budgets
⭐ **Recommended**	**$11.15M**	**51,416 t**	**+$0.00M**	**Balanced**
🌱 **Greenest**	$11.15M	51,416 t	+$0.00M	Emission targets

Why the recommended plan: Delivers 84% of total achievable emissions savings for only a fraction of the cost of the greenest option.

Top Ship Changes

Vessel Identifier	Route Assigned	Key Operational Adjustment
V000	Reserve	no change
V001	Reserve	no change
V002	Reserve	no change
V003	Reserve	no change
V004	Reserve	no change

Can You Trust These Numbers?

- ✓ Calibrated against **21,622 real ship reports** from official European maritime registries (2022–2025).
- ✓ Official **IMO and regulatory emission conversion factors** applied throughout.
- ✓ Optimization rigorously tested against three standard engineering methods: **high-quality schedules, 1.1–1.4× faster**.
- ✓ Fully compliant with international efficiency ratings: 100% of ships achieve A–C rating.

How our method compares

	Our optimizer	Standard methods
Speed	**1.1–1.4× faster than standard method**	Slower
Plans found	Strong compromise solutions (Best in most test runs)	Many near-identical options
Scales to 100 ships	✅	✅
Tested fairly	Same rules, same budget, 5 seeds	✅

What to Watch

📈 **Carbon price threshold:** If carbon prices exceed $85/tonne, green fuels become cheaper than conventional oil in total cost.

📦 **Demand growth:** If cargo volume expands by 15%, the plan easily scales: reserve vessels deploy with only an 11% emissions increase.